STOCK INTELLIGENCE SYSTEMS GROUP

Stock Researcher Agent Master Manual

Integrated Operations: Render REST API, Oracle Cloud OpenClaw, Requesty LLM & Supabase Persistence

Status	PRODUCTION LIVE
Updated	11 October 2026
Repository	appurajacool2015/stock-researcher-agent
Revision	v2.2

Operational purpose

This manual documents the production topology, deployment settings, verification procedures, operational troubleshooting, and planned persistence architecture for the Stock Researcher Agent. It separates deterministic financial data collection and calculations from LLM-generated explanations.

Architectural invariant

> The Render API is a deterministic transport and research layer. The LLM (Requesty/OpenClaw) must not calculate, estimate, or fabricate financial indicators. Numerical metrics, status tags, and provenance must originate from the canonical `ResearchResult v1` payload.

1. Global System Topology & Component Roles

The system isolates deterministic financial computing from natural-language synthesis across four discrete environments.

```
User / Chat Clients (Telegram, WebUI, CLI)
        |
        v
Oracle Cloud VM: 129.225.120.74
  ■■ OpenClaw Gateway Service
     ■■ Health: systemctl --user status openclaw-gateway
     ■■ LLM Inference: Requesty (google/gemma-4-31b-it)
     ■  ■■ Configuration: ~/.openclaw/.env
     ■■ Tool Execution: skills/stock-research/
        ■■ Calls remote Render microservice over HTTPS
           GET /research/{symbol} or POST /research
        |
        v
Render Production Web Service (Singapore)
  ■■ FastAPI App: interfaces.api:app (Uvicorn)
     ■■ Deterministic Pipeline: services.research_pipeline
        ■■ Screener.in Adapter
        ■  ■■ Consolidated fundamentals & shareholding
        ■■ Yahoo Finance Adapter (yfinance)
        ■  ■■ 50/200 DMA, RSI14, volume traction
        ■■ Google News RSS Adapter
           ■■ Article deduplication & clustering
        |
        v
Supabase Cloud Database (PostgreSQL) — ROADMAP
  ■■ stock_research_runs
     ■■ 6-hour caching & historical analytics
```

2. Executive Status Checkpoint

Tier / Component	Environment	Verified Status	Observed Operational Evidence
REST API Microservice	Render (Singapore)	LIVE (200 OK)	stock-researcher-agent.onrender.com running Uvicorn on port 10000; health and TCS verified.
Agent Orchestrator	Oracle Cloud Ubuntu VM	ACTIVE	OpenClaw gateway running under systemd; reachable via SSH key.
Language Model	Requesty Router	CONNECTED	Model google/gemma-4-31b-it verified via OpenClaw CLI response benchmark.
Historical Context	Amazon Bedrock	DEPRECATED	Superseded due to Nova authentication failures. Do not re-enable during recovery.
Research Storage	Supabase PostgreSQL	ROADMAP	Schema defined; planned caching of payloads to reduce redundant scraping.

Architectural invariant: All numerical metrics, status tags, and provenance records must come from the canonical `ResearchResult v1` payload.

3. Render Production Deployment Specification

The REST API is deployed from the primary GitHub repository on Render. The following values document the configuration supplied for the active service.

Parameter	Configured Value	Operational Impact
Repository & Branch	appurajacool2015/stock-researcher-agent (main)	Auto-syncs on git push after local validation.
Region & Runtime	Singapore (Southeast Asia) · Python 3	Latency profile for NSE market scrapers.
Root Directory	Blank / repository root	Ensures interfaces/ and services/ modules resolve.
Build Command	pip install -r requirements.txt	Installs FastAPI, Uvicorn, yfinance, bs4, requests.
Start Command	uvicorn interfaces.api:app --host 0.0.0.0 --port $PORT	Binds to Render's allocated container port.
Health Check Route	/health	Avoids root-path HEAD / 404 warnings in deployment logs.

4. Live API Verification & Evidence Records

Both primary routes were reported as verified with curl immediately after deployment commit `7ef7c63`.

4.1 Health verification

```
curl -i https://stock-researcher-agent.onrender.com/health

HTTP/2 200 OK
{"ok": true, "service": "stock-researcher-api", "status": "healthy"}
```

4.2 Live research execution

```
curl -i https://stock-researcher-agent.onrender.com/research/TCS

HTTP/2 200 OK
{
  "schema_version": "1.0",
  "symbol": "TCS",
  "exchange": "NSE",
  "company_name": "Tata Consultancy Services Ltd",
  "symbol_match_confidence": "EXACT",
  "overall_status": "PARTIAL",
  "fundamentals": {
    "pe_ratio": {"value": 14.3, "status": "VERIFIED"},
    "roce": {"value": 63.0}
  },
  "technicals": {
    "cmp": {"value": 2156.0},
    "rsi_14": {"value": 50.6},
    "volume_traction": true
  },
  "data_quality_summary": {
    "verified": 25, "derived": 4, "unavailable": 2,
    "stale": 0, "conflicting": 0, "failed": 0
  }
}
```

Interpreting PARTIAL status and free-tier latency

• PARTIAL: For zero-debt firms such as TCS, Screener may not report debt-to-equity, and promoter pledge may be absent. These fields can be recorded as `UNAVAILABLE`. A PARTIAL result can be valid and should not automatically be treated as a pipeline fault.

• Cold start: On the Free instance plan, inactivity may spin the container down. The first request after roughly 15 minutes idle was observed to take 40–55 seconds; subsequent requests were observed to complete in 2–15 seconds. Treat these as operational observations, not service-level guarantees.

5. OpenClaw Setup on Oracle Cloud VM

The OpenClaw gateway runs on an OCI Ubuntu instance and brokers chat applications to the deterministic Render backend.

5.A Connectivity & gateway health check

```
# 1. Connect to the Oracle VM
ssh -i ssh-key-2026-08-27.key ubuntu@129.225.120.74

# 2. Verify OpenClaw gateway status
systemctl --user status openclaw-gateway --no-pager

# 3. Test Requesty LLM connection using the validated model
openclaw models list
openclaw agent --agent main --message "Reply with exactly: Requesty connection OK." --timeout 60

# 4. Confirm VM egress reachability to Render
curl -fsS https://stock-researcher-agent.onrender.com/health
```

5.B Wiring the stock-research skill

Update `skills/stock-research/SKILL.md` on the VM so the agent calls the Render REST API rather than local subprocesses.

```
import requests

def execute_stock_research(symbol: str) -> dict:
    normalized = symbol.strip().upper()
    url = f"https://stock-researcher-agent.onrender.com/research/{normalized}"
    response = requests.get(url, timeout=75)  # Allows for cold starts
    response.raise_for_status()
    return response.json()
```

Security note: validate or normalize the symbol before interpolating it into the URL; use URL encoding or a query/body route where appropriate, and do not log secrets.

6. Operational Troubleshooting Matrix

Observed Symptom	Probable Root Cause	Resolution Procedure
Client timeout (>50s)	Render Free instance cold start during initial boot.	Increase OpenClaw tool timeout to 75s; retry and inspect Render deployment logs.
Render logs show 404	Probe is requesting GET / instead of /health.	Set Health Check Path to /health in Render Settings → Web Service.
Symbol error (400)	Input contains invalid special characters such as $, spaces.	Sanitize with normalize_symbol(); allow only supported letters, numbers, dot, and dash.
Requesty failure	API key missing or expired in ~/.openclaw/.env.	Verify REQUESTY_API_KEY is configured. Never print key values in chat or logs.

7. Upcoming Phase: Supabase Persistence Architecture

Supabase (PostgreSQL) is planned as a shared persistence layer to reduce redundant scraping, preserve research history, and support portfolio analytics across chat sessions.

7.1 Migration / snapshot table

```
CREATE TABLE IF NOT EXISTS stock_research_runs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    symbol VARCHAR(20) NOT NULL,
    exchange VARCHAR(10) DEFAULT 'NSE',
    company_name TEXT,
    overall_status VARCHAR(20) NOT NULL,
    data_quality_summary JSONB NOT NULL,
    result_payload JSONB NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_stock_runs_symbol_created
    ON stock_research_runs (symbol, created_at DESC);

CREATE INDEX idx_stock_runs_status
    ON stock_research_runs (overall_status);
```

7.2 Read-cache flow

Step	Operation
1	A chat request arrives for a stock symbol.
2	Check Supabase for a result for that symbol from the last six hours.
3	Cache hit: return cached result_payload without scraping.
4	Cache miss: call Render API, persist the result in Supabase, then reply.

7.3 Security & connection

- Store SUPABASE_URL and SUPABASE_KEY in environment configuration on Oracle VM and Render.
- Enable Row Level Security (RLS) on `stock_research_runs`.
- Keep service-role credentials only in private backend workers; never expose them to clients.

8. Developer Cheatsheet & Local Environment Reference

Task / Operation	Standard Command / Path	Expected Outcome
Local Tests	.venv-market/bin/python -m unittest tests/test_api.py	Reported: 12 tests OK (0.02s).
Run Local Server	.venv-market/bin/python -m interfaces.api	Uvicorn on http://0.0.0.0:8000.
Git Push to Deploy	git pull --rebase origin main && git push origin main	Triggers Render auto-deploy.
Interactive Docs	https://stock-researcher-agent.onrender.com/docs	FastAPI Swagger UI Explorer.

Fast verification one-liner

```
curl -s https://stock-researcher-agent.onrender.com/research/INFY | jq \
  '{symbol: .symbol, status: .overall_status, cmp: .technicals.cmp.value, \
    quality: .data_quality_summary}'
```

End of Master Manual · Revision v2.2 · 11 October 2026

Stock Researcher Agent

Account Directory & Current Development / Runtime Framework

Updated: 11 October 2026 · Companion addendum for the operational documents

1. Account directory

Platform / purpose	Account identifier / notes
Oracle Cloud VM	Account label: ajjw2025 (Root user, as supplied). VM: 129.225.120.74. Existing SSH command uses Linux user ubuntu; this is distinct from the OCI account label.
Render	ajjw2025@gmail.com — deployment dashboard, service settings, logs, and production service.
Requesty API	ajjw2025@gmail.com — model/API account. Base URL: https://router.requesty.ai/v1. Model: google/gemma-4-31b-it.
GitHub / source repository	appurajacool2015@gmail.com — repository owner/account for appurajacool2015/stock-researcher-agent.
ChatGPT setup notes	<link href="https://chatgpt.com/share/6acb63bd-5eb0-83ec-8f20-d573edb9ace1" color="#276FBF">Open shared ChatGPT setup discussion</link> — setup questions and decisions.
Gemini setup notes	<link href="https://gemini.google.com/app/a4890209f82bbe83" color="#276FBF">Open Gemini setup discussion</link> — setup questions and decisions.

Account emails and labels are identifiers, not passwords or API keys. Keep API-key values, passwords, recovery codes, and .env contents out of documentation, Git, screenshots, chat, and logs.

2. Current development framework

Stage	Current workflow
Local development	Use Antigravity as the local development environment to edit and validate the Stock Researcher Agent code.
Source control	Push approved code to the GitHub repository appurajacool2015/stock-researcher-agent using the GitHub account appurajacool2015@gmail.com.
Deployment	Render is connected to the repository/branch and deploys the API from the source code. The documented production service is https://stock-researcher-agent.onrender.com.

3. Current user request and runtime flow



Request path: a user asks for a stock through Telegram or WhatsApp; OpenClaw handles the conversation and uses Requesty for LLM inference. The stock-research skill/tool should call the Render HTTPS API, receive the deterministic ResearchResult payload, and format the response for the user.

Important architecture distinction

GitHub is the code repository and deployment source; it should not be treated as the live market-data endpoint. Based on the existing operational evidence, Render is the live research API. If “github.io” means a separately configured GitHub Pages site or a data-publishing endpoint, that is not evidenced in the supplied operational documents and would need to be documented and verified separately.

4. Current service references

Component	Reference
Render health	https://stock-researcher-agent.onrender.com/health
Render research example	https://stock-researcher-agent.onrender.com/research/TCS
FastAPI interactive docs	https://stock-researcher-agent.onrender.com/docs
Requesty base URL	https://router.requesty.ai/v1
OpenClaw gateway status	systemctl --user status openclaw-gateway --no-pager

Operational guardrail: numerical metrics, status tags, and provenance must come from the deterministic research payload. The LLM must not invent or independently calculate financial indicators.

Master Manual — Command Correction

Revision v2.4 · Correction to Section 8: Developer Cheatsheet

Why the previous command failed

The command in the earlier PDF used line-continuation backslashes inside the jq filter. When copied with formatting or non-breaking spaces, a literal backslash can reach jq and cause an INVALID_CHARACTER / shell quoting error. Markdown link syntax such as [URL](URL) must also not be pasted into the terminal.

Use this single-line command

```
curl -fsS 'https://stock-researcher-agent.onrender.com/research/INFY' | jq '{symbol: .symbol, status: .overall
```

What it does

Calls the Render research endpoint for INFY, parses the JSON response with jq, and prints the symbol, overall research status, CMP field, and data-quality summary.

If jq is not installed

```
curl -fsS 'https://stock-researcher-agent.onrender.com/research/INFY'
```

This returns the full JSON response without filtering. If the request fails, inspect the HTTP/network error and Render logs; this command correction does not itself verify current API availability.